

The background features a dark blue gradient with a subtle pattern of white stars and constellations. Overlaid on this are several technical diagrams in a lighter blue color. These include circular gauges with radial scales and tick marks, some with numbers like 150, 160, 170, 190, 200, 210, 220, 230, 240, 250, and 260. There are also circular arrows indicating clockwise or counter-clockwise rotation, and dashed lines representing paths or orbits.

2. MECHANICAL PROPERTIES OF FLUIDS

PART I

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RECAPITULATION

In XIth Std. we discussed the behaviour of solids under the action of a force. Let us see the difference between solid, liquid and gas ...

	Properties	Solids	Liquids	Gases
1	Mass	Definite	Definite	Definite
2	Shape	Definite	Acquires the shape of the container	Acquires the shape of the container
3	Volume	Definite	Definite	Indefinite
4	Compressibility	Not possible	Almost Negligible	Highly Compressible
5	Fluidity	Not possible	Can flow	Can flow
6	Rigidity	Highly rigid	Less rigid	Not rigid
7	Diffusion	Slow	Fast	Very fast
8	Space between particles	Most closely packed 	Less closely packed 	Least closely packed 
9	Interparticle force	strongest	Slightly weaker than	Negligible

INTRODUCTION

- A fluid means a substance that can flow. Therefore, liquids and gases, collectively, are called fluids.
- A fluid either has no rigidity or its rigidity is very low.
- In our daily life, we often experience the pressure exerted by a fluid at rest and in motion. Viscosity and surface tension play an important role in nature. We will try to understand such properties in this chapter



FLUID

- Any substance that can flow is a fluid.
- A fluid is a substance that deforms continually under the action of an external force.
- Fluid is a phase of matter that includes liquids, gases and plasmas.



MECHANICAL PROPERTIES OF FLUID

An ideal fluid has the following properties:

1. It is incompressible: its density is constant.
2. Its flow is irrotational: its flow is smooth, there are no turbulences in the flow.
3. It is nonviscous: there is no internal friction in the flow, i.e., the fluid has no viscosity.
4. Its flow is steady: its velocity at each point is constant in time.



DIFFERENCE BETWEEN SOLID AND FLUID

It is important to understand the difference between a solid and a fluid.

Solid

- Solids can be subjected to shear stress (tangential stress) and normal stress, as shown in Figure

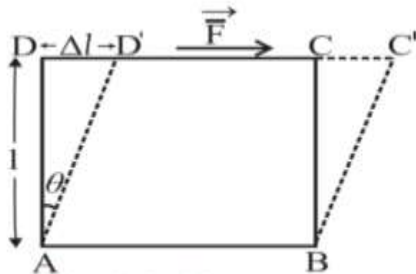


Fig. 2.1: Shear stress.

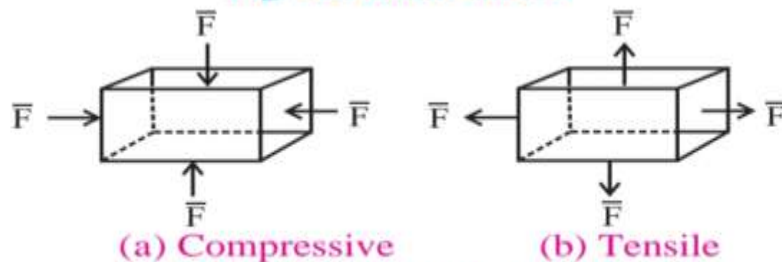


Fig. 2.2: Normal stress.

Fluid

- Ideal fluids, on the other hand, can only be subjected to normal, compressive stress (called pressure) as shown in Figure.

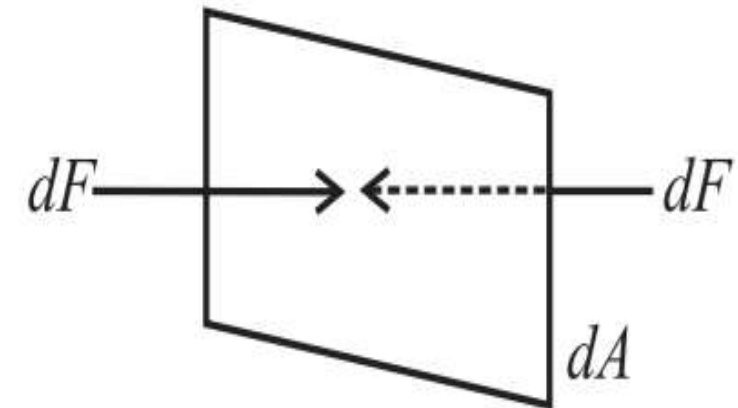


Fig. 2.3: Forces acting on a small surface dA within a fluid at rest.

- Solids opposes the shear stress either by developing a restoring force, which means that the deformations are reversible, or they require a certain initial stress before they deform and start flowing



- Real fluids display viscosity and so are capable of being subjected to low levels of shear stress.



PROPERTIES OF FLUIDS:

1. They do not oppose deformation, they get permanently deformed.
2. They have ability to flow.
3. They have ability to take the shape of the container.

A fluid exhibits these properties because it cannot oppose a shear stress when in static equilibrium



REMEMBER THIS!!!

The term fluid includes both the liquid and gas phases. It is commonly used, as a synonym for liquid only, without any reference to gas. For example, ‘brake fluid’ is hydraulic oil and will not perform its required function if there is gas in it! This colloquial use of the term is also common in the fields of medicine and nutrition, e.g., “take plenty of fluids”

FLUIDS AT REST

The branch of physics which deals with the properties of fluids at rest is called hydrostatics. In the next few sections we will consider some of the properties of fluids at rest



PRESSURE

- A fluid at rest exerts a force on the surface of contact. The surface may be a wall or the bottom of an open container of the fluid.
- The normal force (F) exerted by a fluid at rest per unit surface area (A) of contact is called the pressure (p) of the fluid.

$$p = F/A$$

Figure 2.4 shows a fluid exerting normal force on a vertical surface and Fig. 2.5 shows fluid exerting normal force on a horizontal surface.

surface.

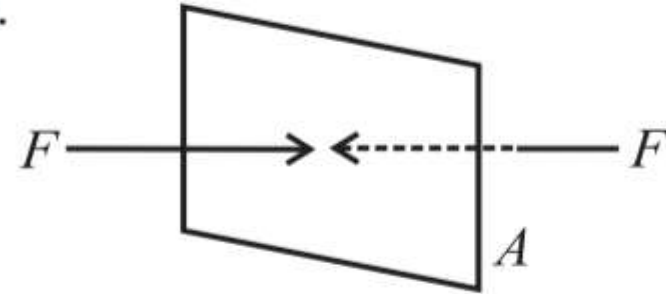


Fig. 2.4: Fluid exert force on vertical surface.

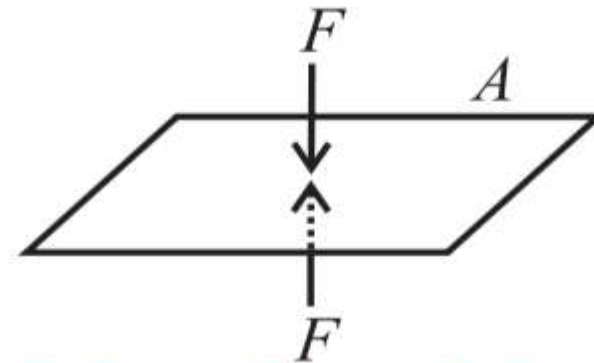


Fig. 2.5: Fluid exert force on horizontal surface.

Thus, an object having small weight can exert high pressure if its weight acts on a small surface area.

REMEMBER THIS!!!

- 1 N weight is about 100 g mass, if $g = 10 \text{ ms}^{-2}$.
- The SI unit of pressure is N/m^2 Also, $1 \text{ N/m}^2 = 1 \text{ Pascal (Pa)}$.
- The dimension of pressure is $[\text{L}^{-1}\text{M}^1\text{T}^{-2}]$.
- Pressure is a scalar quantity.
- Other common units of measuring pressure of a gas are bar and torr.
- $1 \text{ bar} = 10^5 \text{ Nm}^{-2}$, $1 \text{ hectapascal (hPa)} = 100 \text{ Pa}$

REMEMBER THIS!!!

The concept of pressure is useful in dealing with fluids, i.e., liquids and gases. As fluids do not have definite shape and volume, it is convenient to use the quantities pressure and density rather than force and mass when studying hydrostatics and hydrodynamics.

PRESSURE DUE TO A LIQUID COLUMN

- Consider a vessel is filled with a liquid.
- Let us calculate the pressure exerted by an imaginary cylinder of cross sectional area A inside the container.
- Let the density of the fluid be ρ , and the height of the imaginary cylinder be h as shown in the Figure.
- The liquid column exerts a force

$$F = mg$$

which is its weight, on the bottom of the cylinder.

- This force acts in the downward direction. Therefore, the pressure p exerted by the liquid column on the bottom of cylinder is,

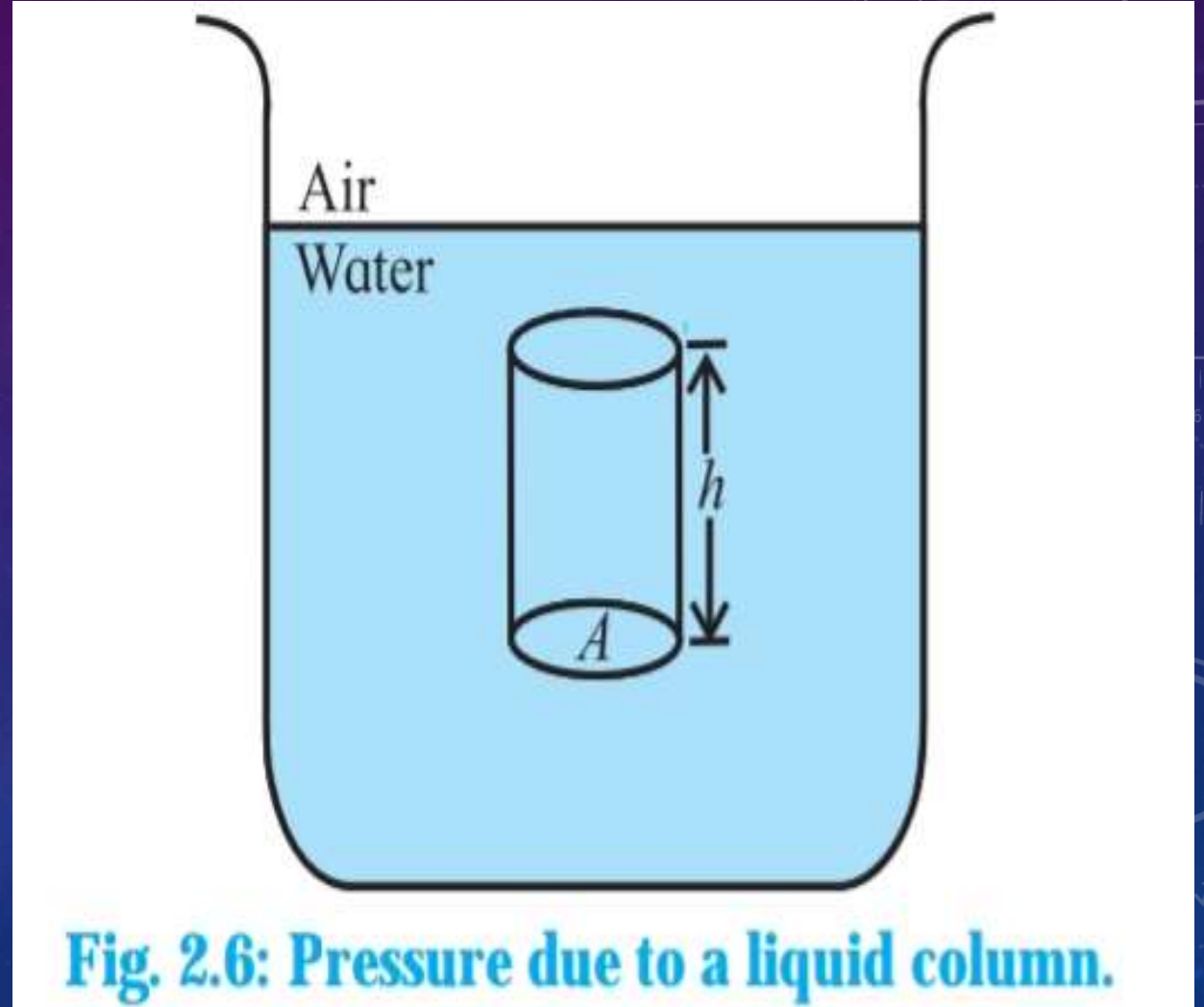


Fig. 2.6: Pressure due to a liquid column.

$$p = \frac{F}{A}$$

$$\therefore p = \frac{mg}{A}$$

Now, $m = (\text{volume of cylinder}) \times (\text{density of liquid})$

$$= (Ah) \times \rho = Ah\rho$$

$$\therefore p = \frac{(Ah\rho)g}{A}$$

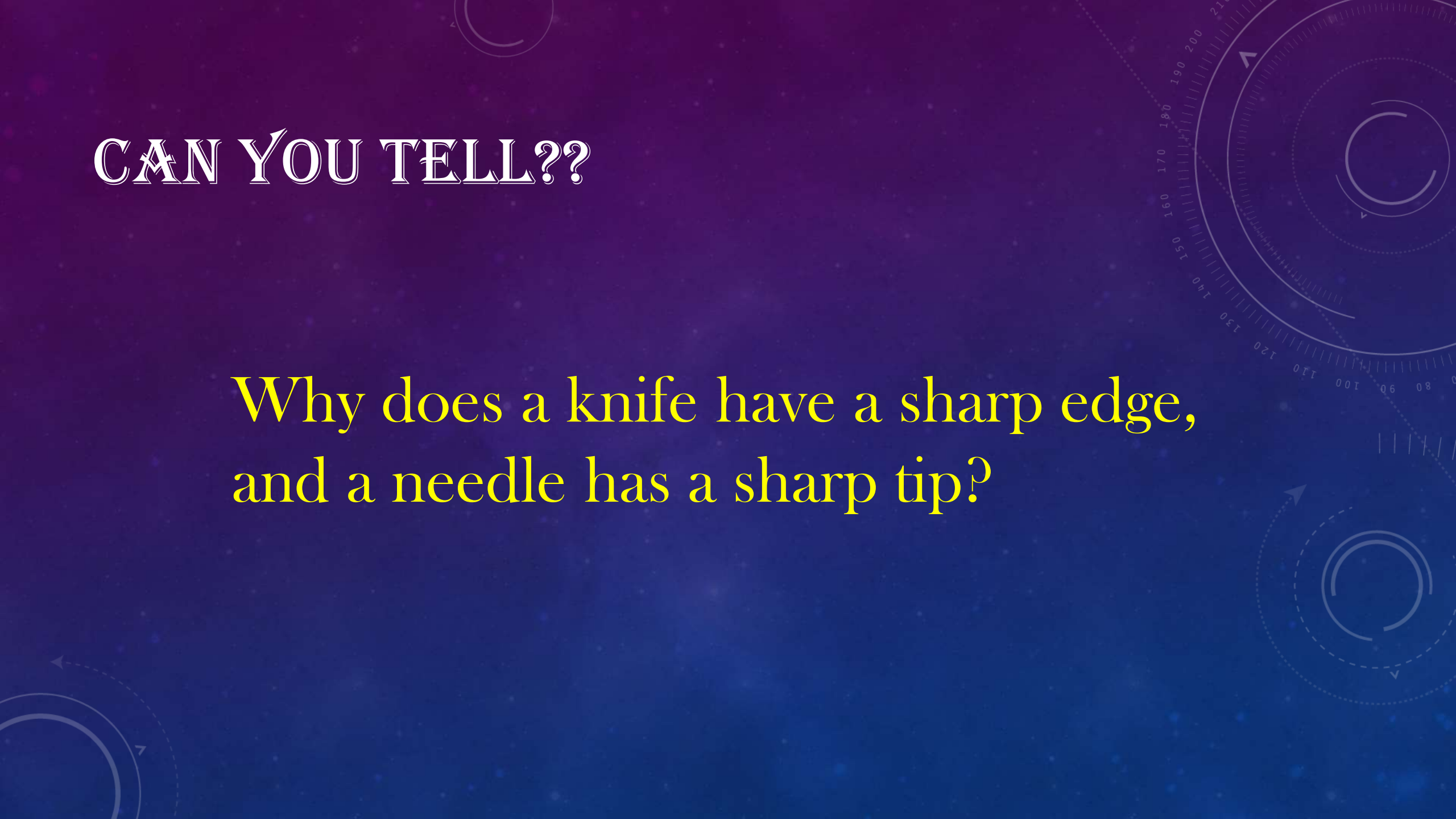
$$p = h\rho g \quad \text{--- (2.2)}$$

Thus, the pressure p due to a liquid of density ρ at rest, and at a depth h below the free surface is $h\rho g$.

Note that the pressure does not depend on the area of the imaginary cylinder used to derive the expression

REMEMBER THIS!!!

- As $p = h\rho g$, the pressure exerted by a fluid at rest is independent of the shape and size of the container.
- $p = h\rho g$ is true for liquids as well as for gases

The background is a dark blue gradient with faint, light blue technical diagrams. These include circular gauges with scales (one labeled 100-210, another 130-160), concentric circles, and dashed lines with arrows indicating motion or flow. The overall aesthetic is scientific or engineering-related.

CAN YOU TELL??

Why does a knife have a sharp edge,
and a needle has a sharp tip?

ANSWER!!!

The cutting edge of tools like blades, knives etc., are provided with sharp edge to cut the things easily as the sharp edges have small areas on which the force is applied has more pressure applied.

USE YOUR BRAIN POWER!!!

A student of mass 50 kg is standing on both feet. Estimate the pressure exerted by the student on the Earth. Assume reasonable value to any other quantity you need. Justify your assumption. You may use $g = 10 \text{ m s}^{-2}$

By what factor will it change if the student lies on back?

SOLUTION

The pressure exerted on the earth is 25 Pa.

EXPLANATION

Explanation: Mass of student = 50 Kg

Gravitational acceleration "g" = 10 m/s^2

- Solution: According to newton's third law of motion.

$$F = ma$$

$$F = (50 \times 10) \text{ N} = 500 \text{ N}$$

Let the Area be 20 m^2 .

We know that

- $\text{Pressure} = \text{Force} / \text{Area}$

- $\text{Pressure} = (500 / 20)$

$$\text{Pressuree "P"} = 25 \text{ Pa}$$

Thuss the pressure exerted on the earth is 25 Pa.

REVIEW

Fluid

- Any substance that can flow is a fluid. A fluid is a substance that deforms continually under the action of an external force.
- It is incompressible
- Its flow is irrotational
- It is nonviscous
- Its flow is steady

Pressure due to a liquid column

- As $p = h\rho g$, the pressure exerted by a fluid at rest is independent of the shape and size of the container.

Pressure

- Pressure is defined as Force per unit area
- The SI unit of pressure is N/m^2
- $1 \text{ N/m}^2 = 1 \text{ Pascal (Pa)}$.
- The dimension of pressure is $[\text{L}^{-1}\text{M}^1\text{T}^{-2}]$.
- Pressure is a scalar quantity.
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- $1 \text{ bar} = 10^5 \text{ Nm}^{-2}$
- $1 \text{ hectapascal (hPa)} = 100 \text{ Pa}$

THANKS

**WE NEVER KNOW
THE WORTH OF
WATER TILL
THE WELL
IS DRY.**

FRENCH PROVERB